Pose-coupled native-interaction weighting for kinase ensemble docking: retrospective evaluation on EGFR and CDK2

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1 Abstract

Background. Docking and neural rescoring can rank ligands without establishing whether the selected receptor-specific pose recovers interactions observed in target holo structures. We asked whether a target-native interaction prior adds incremental early-enrichment information to GNINA when both signals are evaluated on the same pose.

Methods. Syndesis combined five-state EGFR ensemble docking, GNINA CNNscore, and ProLIF interaction fingerprints. A native-derived ATP-site union prior was constructed from four EGFR holo complexes. The primary ligand score was the maximum, across receptor states, of CNNscore multiplied by one plus same-pose interaction recall. We used paired class-stratified bootstraps, three permutation controls, receptor and native-complex exclusions, exact native-ligand-overlap audits, and similarity and size analyses. CDK2 provided a transfer boundary analysis.

Results. Pose-coupled weighting increased EGFR EF1% from 11.79 to 16.40, recovering 89 rather than 64 actives among the first 356 ranked molecules. The paired EF1% difference was 4.61 (95% CI 2.58 to 6.82). The effect exceeded unrestricted, heavy-atom-count-matched, and class-conditional assignment nulls; it persisted after every receptor exclusion, removal of the exact-overlap native ligand, and joint removal of duplicate AQ4 complexes. CDK2 showed a favorable but unresolved EF1% difference of 2.11 (95% CI −0.42 to 4.64).

Conclusions. A target-native interaction prior can improve EGFR early enrichment when interaction recall and neural score are coupled to the same receptor-specific pose. The result is target- and ensemble-dependent rather than evidence for a universal rescoring rule.

Scientific Contribution. We introduce and evaluate a pose-coupled native-interaction weighting rule that exposes the structural evidence behind a neural docking score. Paired and permutation controls distinguish its EGFR early-enrichment signal from score-combination, molecular-size, class-distribution, receptor, and native-ligand-overlap effects.

Keywords: structure-based virtual screening; docking; interaction fingerprints; early enrichment; EGFR; CDK2

2 Background

Structure-based virtual screening is a ranking problem conditioned on pose generation. A docking program may sample a near-native geometry without ranking it first, and a favorable score alone does not establish that the selected pose is structurally credible (Trott and Olson 2010; Amaro et al. 2018; Buttenschoen et al. 2024). Neural scoring functions such as GNINA improve pose assessment by learning protein-ligand patterns from three-dimensional data, but their outputs remain statistical scores rather than explicit tests of whether a pose retains contacts characteristic of a target (McNutt et al. 2021, 2025).

Protein-ligand interaction fingerprints provide that explicit representation. SIFt, SPLIF, kinase interaction profiles, and tools such as ProLIF encode complexes as residue-by-interaction descriptors and have been used for pose comparison, binding-mode analysis, and rescoring (Deng et al. 2004; Chuaqui et al. 2005; Marcou and Rognan 2007; Da and Kireev 2014; Bouysset and Fiorucci 2021). Prior fingerprint-based methods commonly compare a docked pose with a reference pattern after docking. The unresolved methodological issue is how to combine this reference evidence with a learned score across a receptor ensemble.

That combination must preserve pose identity. If the highest neural score is taken from one receptor state and the highest interaction value from another, their combination can reward two incompatible poses. The resulting late-fusion score is not evidence for any single protein-ligand geometry and may gain enrichment mechanically as the number of receptor states grows. We instead evaluate both terms for the same ligand, receptor state, and docked pose before maximizing over receptor states.

Kinase ATP pockets provide a stringent test because they contain recurrent hinge and catalytic interactions while sampling multiple conformational states and diverse ligand chemotypes. We therefore tested one narrow hypothesis: **does a native-derived interaction prior add early-enrichment information beyond GNINA when the two quantities are computed from the same receptor-specific pose?** EGFR was the method-development and retrospective evaluation target; CDK2 was used to define transfer limits. The downstream Syndesis workflow retains pose, analog, and MD evidence for inspection, but those layers are not presented as activity prediction.

3 Methods

3.1 Study design and structural inputs

The primary study was a paired comparison between GNINA and a pose-coupled GNINA-plus-interaction score on the same EGFR ligand-receptor evaluations. The EGFR docking ensemble contained 1M17, 1XKK, 4HJO, 5CAV, and 6DUK. The interaction prior used only ATP-site holo complexes 1M17/AQ4, 1XKK/FMM, 4HJO/AQ4, and 5CAV/4ZQ; the allosteric ligand in 6DUK was excluded from prior construction (To et al. 2019). CDK2 used 1QMZ/ATP, 1FIN/ATP, 2A4L/RRC, 1AQ1/STU, and 1PXN/CK6; 1H00 was held out. Receptor identities, chains, docking boxes, quality decisions, and residue maps are machine-readable in the release.

EGFR receptor chains were prepared from PDB structures (Berman et al. 2000), with non-protein residues removed and Open Babel 3.1.0 used to produce pH 7.4, Gasteiger-charged PDBQT receptors. For docked-pose fingerprints, the ProLIF protein was regenerated from the exact docking PDBQT by Open Babel; ProLIF then assigned residue, aromatic, donor, and acceptor chemical perception to that docking-derived model. A five-receptor audit found identical docking and ProLIF heavy-atom sets and zero ProLIF-only atoms. The CDK2 chain-A extractor represented 1QMZ without HETATM phosphothreonine TPO160; the 1QMZ-excluded result is therefore treated as a sensitivity analysis rather than a phosphorylated-CDK2 model.

Table 1. Receptor ensembles, benchmark sizes, and native-prior complexes.

Target	Docking receptor states	Benchmark molecules	Native-prior complexes
EGFR	1M17, 1XKK, 4HJO, 5CAV, 6DUK	35,552 (542 actives; 35,010 decoys)	4 ATP-site complexes
CDK2	1QMZ, 1FIN, 2A4L, 1AQ1, 1PXN	28,296 (474 actives; 27,822 decoys)	5

3.2 Docking, interaction encoding, and score coupling

DUD-E input SMILES were converted to one ETKDGv3/MMFF94-optimized three-dimensional state per record and then to PDBQT with Open Babel (Mysinger et al. 2012; Landrum 2013; Halgren 1996; O'Boyle et al. 2011). Alternative protomers, tautomers, stereoisomers, and conformers were not enumerated. Each ligand was docked independently to all five receptor states with Uni-Dock 1.2.0 in `balance` mode, nine output modes, and seed 807 (Yu et al. 2023). Only the top Uni-Dock pose per ligand-receptor pair entered the enrichment analysis; GNINA 1.3.3 rescored that pose (McNutt et al. 2021, 2025).

ProLIF 2.2.0 encoded hydrophobic, implicit donor and acceptor, ionic, cation-pi, pi-cation, and van der Waals interactions as normalized residue-by-interaction-type bits (Bouysset and Fiorucci 2021). Docked coordinates were transferred onto the prepared SDF graph before fingerprinting, preserving bond order, formal charge, tautomerism, and stereochemistry. Atom-order, element, count, and coordinate-mapping failures were not converted to zero scores; all 177,760 EGFR ligand-receptor evaluations passed this reconstruction audit.

Let $F_{i,r}$ be the fingerprint for ligand i in receptor state r, and N_k the fingerprint for native complex k. The primary prior was the target-native union $C = \bigcup_k N_k$, with 62 EGFR and 47 CDK2 bits. Same-pose recall was $R_{i,r} = \mid F_{i,r} \cap C \mid / \mid C \mid$. The coupled ligand score was

$$S_i = \max_r \left\{ \text{CNNscore}_{i,r}[1 + R_{i,r}] \right\}.$$

The multiplier is bounded between one and two. Critically, CNNscore and recall in each product belong to the same pose. EGFR labels informed development, so this is method development plus retrospective evaluation; the formula, $\lambda = 1$, EF1% endpoint, bootstrap design, and seeds were frozen before final strict fingerprint recomputation. Alternative priors and formulas were sensitivity analyses, not selection criteria.

3.3 Statistical evaluation

EF1% was the primary endpoint; ROC-AUC, EF5%, and BEDROC ($\alpha = 80.5$) were secondary (Truchon and Bayly 2007). We used 2,000 paired class-stratified bootstrap resamples (seed 807). Three 1,000-draw permutation controls reassigned complete five-receptor recall vectors: across all ligands, within heavy-atom-count deciles, and within activity class. The last preserves active-decoy recall distributions and tests molecule-specific assignment rather than a general random-score null. Receptor exclusions, native-complex exclusions, joint duplicate-chemotype exclusions, exact native/DUD-E identity checks, ECFP4 similarity strata, and recall-size correlations tested robustness (Bemis and Murcko 1996; Rogers and Hahn 2010). DUD-E decoys are property-matched benchmark compounds rather than experimentally confirmed inactives, and its analogue and decoy construction can introduce benchmark-specific bias; this analysis therefore evaluates incremental ranking performance, not prospective activity prediction (Mysinger et al. 2012; Stein et al. 2021; Wallach and Heifets 2018).

4 Results

4.1 Redocking motivates pose-coupled evidence

Redocking separated pose sampling from pose selection. In one AQ4 task, a 0.83 A pose was sampled while the top-ranked docking pose was 5.81 A from the reference. Across 12 tasks containing a native-like sampled pose, GNINA CNNscore achieved NDCG@1 of 0.833 (95% CI 0.583 to 1.000), compared with 0.500 (0.250 to 0.750) for docking score. This does not validate the enrichment rule, but establishes why evidence attached to the selected pose matters.

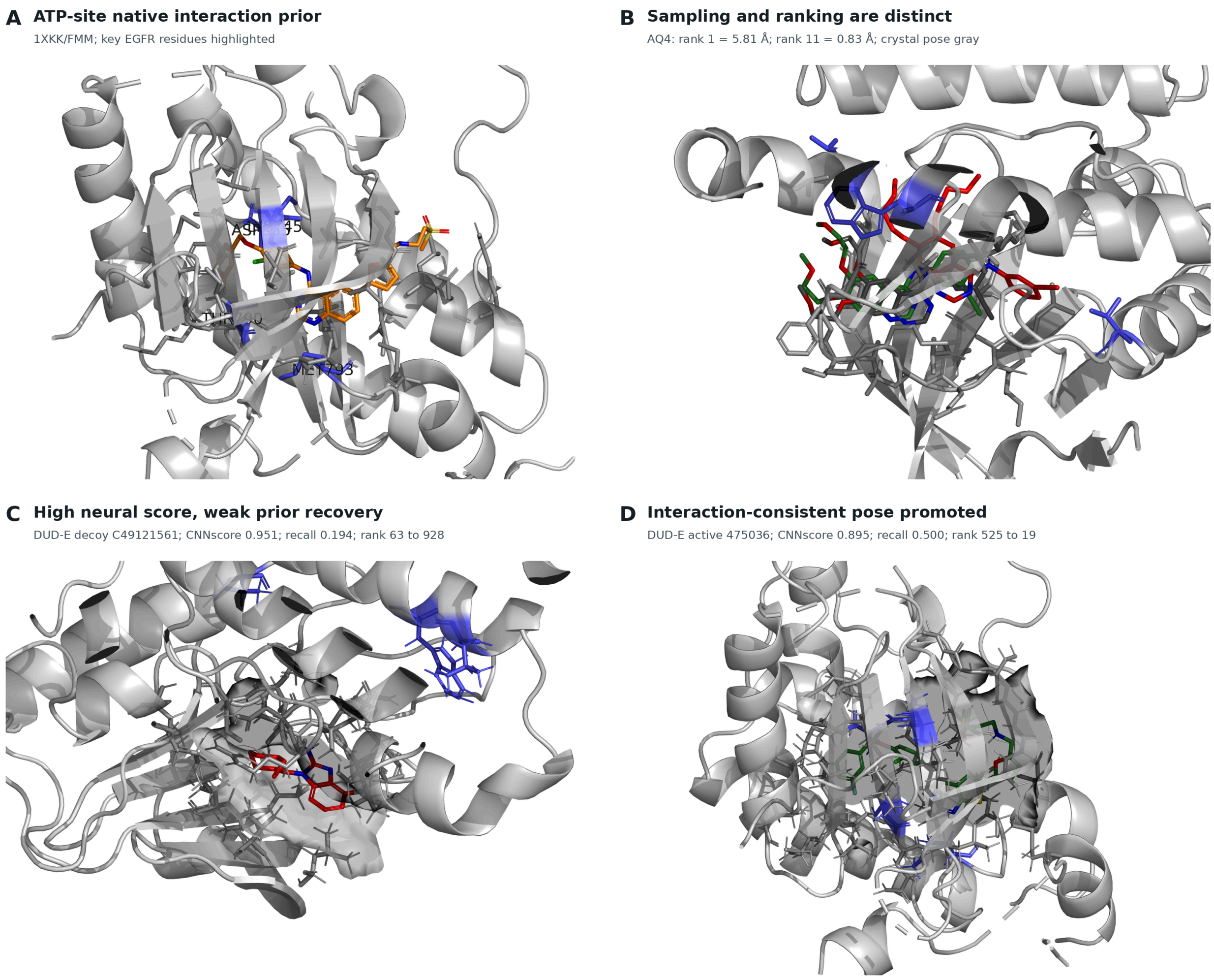

Figure 1: Structural basis of pose-coupled interaction weighting. Native contacts define the target prior; sampled poses can differ from the docker's rank-1 choice; and the coupled score promotes poses that jointly retain neural and interaction evidence.

4.2 Pose-coupled weighting improves EGFR early enrichment

The EGFR benchmark comprised 542 actives and 35,010 decoys. GNINA achieved EF1% 11.79, retrieving 64 actives among the first 356 molecules. The coupled score achieved EF1% 16.40 and retrieved 89 actives. The paired EF1% improvement was 4.61 (95% CI 2.58 to 6.82); ROC-AUC, EF5%, and BEDROC also increased.

Table 2. EGFR enrichment across the five-receptor ensemble. Intervals are percentile 95% confidence intervals from 2,000 class-stratified bootstrap resamples.

Ranking arm	ROC-AUC (95% CI)	EF1% (95% CI)	EF5% (95% CI)	BEDROC (95% CI)
GNINA	0.766 (0.742-0.789)	11.79 (9.40-14.37)	6.90 (6.12-7.71)	0.209 (0.177-0.242)
Pose-coupled score	**0.772 (0.749-0.796)**	**16.40 (13.63-19.35)**	**7.75 (6.90-8.49)**	**0.282 (0.244-0.319)**

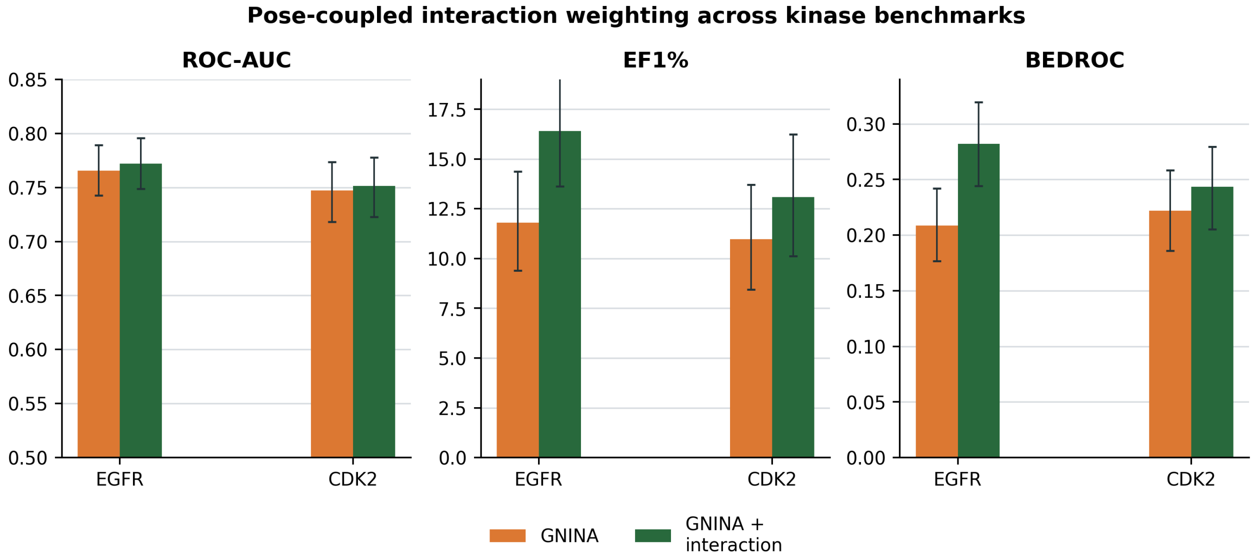

Figure 2: EGFR primary result and CDK2 transfer boundary. Error bars are paired 2,000-bootstrap 95% intervals.

The gain exceeded all three nulls. All-ligand reassignment gave mean EF1% 11.35 (empirical $p = 0.0010$); heavy-atom-count matching gave 12.39 ($p = 0.0010$); and the more stringent class-conditional assignment null gave 14.26 ($p = 0.0040$). Thus, the result is not explained by a generic score combination, ligand-size matching, or active-decoy class distributions alone.

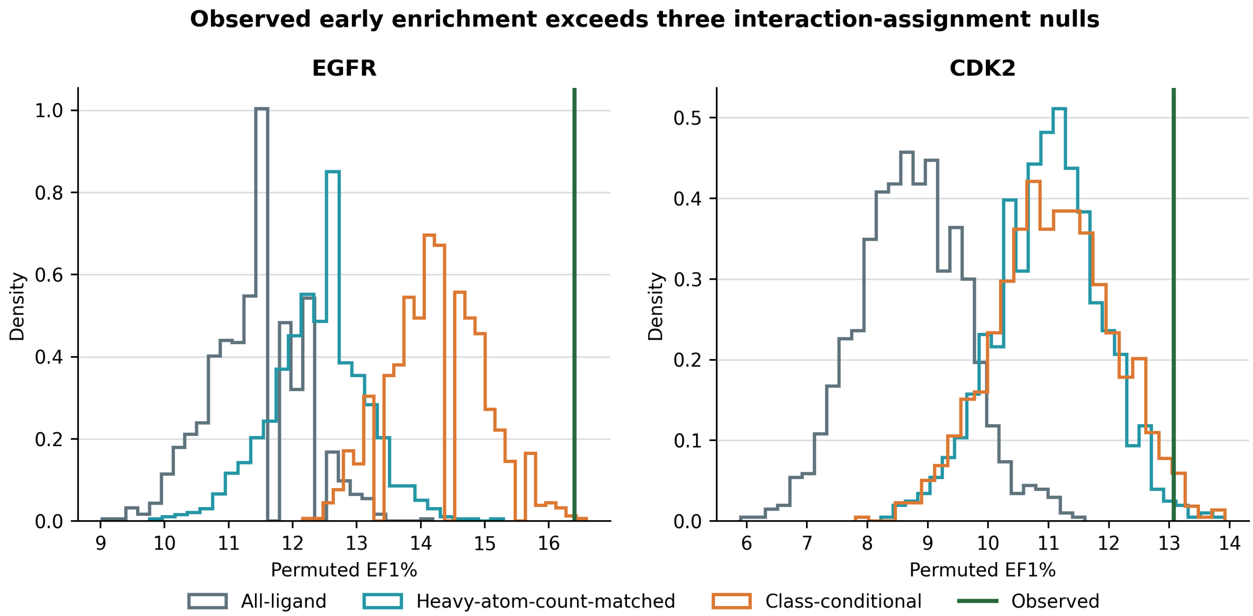

Figure 3: EGFR EF1% under unrestricted, size-matched, and class-conditional interaction-assignment nulls.

4.3 Robustness analyses localize the scope of the EGFR result

No single receptor explained the effect: leave-one-receptor-out EF1% gains ranged from 2.95 to 4.97, with all paired intervals excluding zero. The allosteric 6DUK ligand was absent from the prior in every analysis. Removing the exact-overlap FMM native ligand retained EF1% 15.29 and a paired gain of 3.50 (95% CI 1.84 to 5.53); removing both AQ4 complexes retained EF1% 15.66 and a gain of 3.87 (1.29 to 6.26). Among 369 actives with maximum ECFP4 similarity below 0.30 to every distinct native ligand, the coupled ranking recovered 54 in the global top 1%, compared with 39 for GNINA.

Recall correlated weakly with heavy-atom count ($\rho = 0.158$) and molecular weight ($\rho = 0.160$), but more strongly with total detected contacts ($\rho = 0.569$). This motivates the size-matched null and prevents interpreting union recall as size-free. Conserved-core, frequency-weighted, Jaccard, and receptor-specific priors also improved EGFR enrichment; these are robustness observations, not evidence that one weighting formula is universally optimal.

Across $\lambda = 0.25$ to 3 in $\mathrm{CNNscore}[1 + \lambda R]$, the EGFR direction remained positive; $\lambda = 1$ was the development-fixed primary value rather than the empirically optimal value.

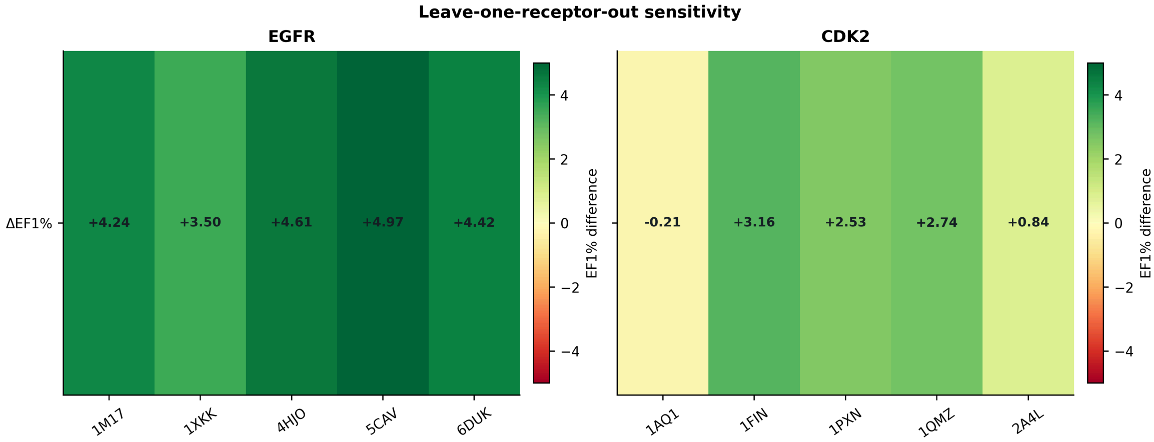

Figure 4: Leave-one-receptor-out EF1% effects for EGFR and CDK2.

4.4 CDK2 defines a transfer boundary

Applying the same procedure to CDK2 increased EF1% from 10.97 to 13.08, corresponding to 52 versus 62 actives among the first 283 molecules. The paired difference was 2.11 (95% CI −0.42 to 4.64), and receptor exclusions were heterogeneous. Its all-ligand, heavy-atom-count-matched, and class-conditional permutation tests gave empirical $p =$ 0.0010, 0.0140, and 0.0220, respectively. The positive direction is compatible with useful interaction information, but the paired uncertainty does not establish transfer beyond the EGFR ensemble. Excluding the unphosphorylated 1QMZ representation gave a difference of 2.74 (0.00 to 5.06), which is a sensitivity result rather than validation of phosphorylated CDK2 chemistry.

5 Discussion

The main finding is methodological and specific: a native-derived interaction prior improved EGFR early enrichment when it was coupled to the GNINA score of the same receptor-specific pose. Same-pose coupling is essential because a receptor ensemble otherwise permits score components from incompatible poses to be combined. The paired comparison and three null designs show that the observed gain is not adequately described as a mechanical consequence of adding another score, preserving class-level recall distributions, or matching ligand size.

The robustness analyses also refine the biological interpretation. The effect survived removal of individual receptor states, exact native-ligand overlap, and duplicate AQ4 structures, and remained visible among low-similarity actives. These results support the view that the prior contributes target-structural information rather than merely recognizing one crystallographic chemotype. At the same time, union recall depends on the number of detected contacts and different prior definitions also perform well. The appropriate conclusion is not that native-union recall is uniquely correct, but that explicit target-native interaction evidence can complement a learned score.

CDK2 sets the boundary of the claim. Its point estimates are favorable, but the paired EF1% interval crosses zero and receptor dependence is substantial. Transfer should therefore be evaluated target by target rather than assumed from kinase-family membership. The one-state ligand-preparation strategy is an additional practical limitation because alternative protonation, tautomeric, stereochemical, and conformational states can affect both docking and interaction recovery. Likewise, DUD-E is external to Syndesis development but not necessarily independent of GNINA training structures or chemistry; this study estimates the incremental value of coupling on these benchmarks, not absolute GNINA generalization.

6 Conclusions

Pose-coupled native-interaction weighting improved EGFR early enrichment beyond GNINA across paired, permutation, receptor-exclusion, native-overlap, and similarity controls. The result supports a focused principle for ensemble docking: interaction evidence should be evaluated on the same pose as the score it modifies. Its unresolved CDK2 transfer result argues for target-specific evaluation, not a universal rescoring claim.

7 Abbreviations

ATP, adenosine triphosphate; BEDROC, Boltzmann-enhanced discrimination of receiver operating characteristic; CDK2, cyclin-dependent kinase 2; CNN, convolutional neural network; DUD-E, Directory of Useful Decoys–Enhanced; ECFP, extended-connectivity fingerprint; EF, enrichment factor; EGFR, epidermal growth factor receptor; IFP, interaction fingerprint; MD, molecular dynamics; NDCG, normalized discounted cumulative gain; ROC-AUC, area under the receiver-operating-characteristic curve.

8 Declarations

8.1 Availability of data and materials

Source code, workflow configurations, tests, figures, the rendered manuscript, and machine-readable supporting data are available at https://github.com/eva-mitropoulou/Syndesis. The exact paper package is the v1.1.3-paper release. It includes pose coordinates, native interaction-bit tables, ligand-level benchmark scores and fingerprints, bootstrap and permutation draws, exclusion analyses, analog lineage, graph-mapping validation, MD protocols and replicate metrics, and prospective ranks. Raw structures and benchmark molecules originate from the PDB, DUD-E, and ZINC22 and remain subject to their source terms. No separate supplementary document accompanies this manuscript.

8.2 Competing interests

The authors declare no competing interests.

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8.4 Authors' contributions

Following the CRediT taxonomy: E.M., conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing–original draft, and writing–review and editing; D.G., conceptualization, methodology, validation, resources, supervision, project administration, and writing–review and editing. Both authors read and approved the manuscript.

8.5 Ethics approval and consent to participate

Not applicable.

8.6 Consent for publication

Not applicable.

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References

Amaro, Rommie E., Jerome Baudry, John Chodera, et al. 2018. "Ensemble Docking in Drug Discovery." *Biophysical Journal* 114 (10): 2271–78. https://doi.org/10.1016/j.bpj.2018.02.038.

Bemis, Guy W., and Mark A. Murcko. 1996. "The Properties of Known Drugs. 1. Molecular Frameworks." *Journal of Medicinal Chemistry* 39 (15): 2887–93. https://doi.org/10.1021/jm9602928.

Berman, Helen M. et al. 2000. "The Protein Data Bank." *Nucleic Acids Research* 28 (1): 235–42. https://doi.org/10.1093/nar/28.1.235.

Bouysset, Cédric, and Stefano Fiorucci. 2021. “ProLIF: A Library to Encode Molecular Interactions as Fingerprints.” *Journal of Cheminformatics* 13: 72. https://doi.org/10.1186/s13321-021-00548-6.

Buttenschoen, Martin, Garrett M. Morris, and Charlotte M. Deane. 2024. “PoseBusters: AI-Based Docking Methods Fail to Generate Physically Valid Poses or Generalise to Novel Sequences.” *Chemical Science* 15: 3130–39. https://doi.org/10.1039/D3SC04185A.

Chuaqui, Claudio, Zhan Deng, and Juswinder Singh. 2005. “Interaction Profiles of Protein Kinase-Inhibitor Complexes and Their Application to Virtual Screening.” *Journal of Medicinal Chemistry* 48 (1): 121–33. https://doi.org/10.1021/jm049312t.

Da, Chao, and Dmitri Kireev. 2014. “Structural Protein-Ligand Interaction Fingerprints (SPLIF) for Structure-Based Virtual Screening: Method and Benchmark Study.” *Journal of Chemical Information and Modeling* 54 (9): 2555–61. https://doi.org/10.1021/ci500319f.

Deng, Zhan, Claudio Chuaqui, and Juswinder Singh. 2004. “Structural Interaction Fingerprint (SIFt): A Novel Method for Analyzing Three-Dimensional Protein-Ligand Binding Interactions.” *Journal of Medicinal Chemistry* 47 (2): 337–44. https://doi.org/10.1021/jm030331x.

Halgren, Thomas A. 1996. “Merck Molecular Force Field. I. Basis, Form, Scope, Parameterization, and Performance of MMFF94.” *Journal of Computational Chemistry* 17 (5-6): 490–519. https://doi.org/10.1002/(SICI)1096-987X(199604)17:5/6<490::AID-JCC1>3.0.CO;2-P.

Landrum, Greg. 2013. “RDKit: Open-Source Cheminformatics.” *GitHub*. https://www.rdkit.org.

Marcou, Gilles, and Didier Rognan. 2007. “Optimizing Fragment and Scaffold Docking by Use of Molecular Interaction Fingerprints.” *Journal of Chemical Information and Modeling* 47 (1): 195–207. https://doi.org/10.1021/ci600342e.

McNutt, Andrew T. et al. 2021. “GNINA 1.0: Molecular Docking with Deep Learning.” *Journal of Cheminformatics* 13: 43. https://doi.org/10.1186/s13321-021-00522-2.

McNutt, Andrew T. et al. 2025. “GNINA 1.3: The Next Increment in Molecular Docking with Deep Learning.” *Journal of Cheminformatics* 17: 28. https://doi.org/10.1186/s13321-025-00973-x.

Mysinger, Michael M. et al. 2012. “Directory of Useful Decoys, Enhanced (DUD-E): Better Ligands and Decoys for Better Benchmarking.” *Journal of Medicinal Chemistry* 55 (14): 6582–94. https://doi.org/10.1021/jm300687e.

O’Boyle, Noel M. et al. 2011. “Open Babel: An Open Chemical Toolbox.” *Journal of Cheminformatics* 3: 33. https://doi.org/10.1186/1758-2946-3-33.

Rogers, David, and Mathew Hahn. 2010. “Extended-Connectivity Fingerprints.” *Journal of Chemical Information and Modeling* 50 (5): 742–54. https://doi.org/10.1021/ci100050t.

Stein, Reed M., Ying Yang, Trent E. Balius, et al. 2021. “Property-Unmatched Decoys in Docking Benchmarks.” *Journal of Chemical Information and Modeling* 61 (2): 699–714. https://doi.org/10.1021/acs.jcim.0c00598.

To, Carlota et al. 2019. “Single and Dual Targeting of Mutant EGFR with an Allosteric Inhibitor.” *Cancer Discovery* 9 (7): 926–43. https://doi.org/10.1158/2159-8290.CD-18-0903.

Trott, Oleg, and Arthur J. Olson. 2010. “AutoDock Vina: Improving the Speed and Accuracy of Docking with a New Scoring Function, Efficient Optimization, and Multithreading.” *Journal of Computational Chemistry* 31 (2): 455–61. https://doi.org/10.1002/jcc.21334.

Truchon, Jean-François, and Christopher I. Bayly. 2007. “Evaluating Virtual Screening Methods: Good and Bad Metrics for the "Early Recognition" Problem.” *Journal of Chemical Information and Modeling* 47 (2): 488–508. https://doi.org/10.1021/ci600426e.

Wallach, Izhar, and Abraham Heifets. 2018. “Most Ligand-Based Classification Benchmarks Reward Memorization Rather Than Generalization.” *Journal of Chemical Information and Modeling* 58 (5): 916–32. https://doi.org/10.1021/acs.jcim.7b00403.

Yu, Yuejiang, Chun Cai, Jiayue Wang, Zonghua Bo, Zhengdan Zhu, and Hang Zheng. 2023. “Uni-Dock: GPU-Accelerated Docking Enables Ultralarge Virtual Screening.” *Journal of Chemical Theory and Computation* 19 (11): 3336–45. https://doi.org/10.1021/acs.jctc.2c01145.